Software Requirements Documentation

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Document Control

Attribute	Value
Document ID	CRM-PRD-001
Product	AAP Fleet Management System (`car_repair_management`)
Version / issue date	1.0 / 13 November 2026
Prepared for	Addis Ababa Police Commission; INSA software production assessment application
Publisher	Selfmade Cloud Solutions
Source baseline	`5974cf7addfef5d5cfc8fde325fc964343557a2f`
Status	Technical review draft; approval and deployment evidence pending
Approvers	Commission product owner and authorized signatory: TO BE CONFIRMED

1. Purpose

Define the business, functional, interface, data and operational requirements for the Commission's fleet and workshop management system. Requirements are traceable to implementation areas and acceptance checks. This document describes the evaluated application and identifies requirements that still need production validation; it does not claim all security or nonfunctional requirements have been met.

The planned deployment is on-premise on the Commission's server, with more than 700 vehicles and more than 500 users. Registered user count is not simultaneous user count. Technical contact: Natnael Tilaye (natnaeltilaye30@gmail.com). These project facts were supplied by the Commission representative; hardware sizing and peak load remain to be validated.

Status vocabulary: Implemented = identifiable source implementation, not full acceptance; Partial = implementation exists with a specified limitation; Required = production requirement whose evidence or implementation is outstanding; Out of scope = not represented as a delivered feature.

2. Business Context

The Commission needs a single record of vehicles, responsibility assignments, reported defects, maintenance work, fuel allocation, operating costs and hardware observations. Workshop staff need work-order progress and parts visibility. Fleet supervisors need vehicle condition, service reminders, inspection history and exception reporting. Finance personnel need traceability from a repair to its commercial and expense records.

The hardware integration uses a device's IMEI as its stable vehicle association. Provider APIs can be previewed before saving, selected as the site's active configuration, and used for mapped-vehicle synchronization. The system also accepts authenticated device pushes. A successful request is distinct from a new tracker measurement.

2.1 Objectives and Measures

Objective	Proposed measure	Acceptance owner
Trace repair decisions and execution	Sample an issue through work order, task, parts and handover records	Workshop supervisor

Objective	Proposed measure	Acceptance owner
Improve fleet visibility	Review vehicle overview, inspection history, open issues and latest sensor timestamps	Fleet manager
Control fuel processes	Reconcile approved refueling with quota and financial evidence	Fuel/finance owner
Validate hardware ingestion	Match a fictional/test IMEI to exactly one vehicle; confirm GPS and fuel conversion	Integration owner
Repeat deployment reliably	Clean Docker installation and upgrade pass schema audit and workflow tests	Infrastructure owner
Protect Commission information	Negative authorization, network, secrets and restore evidence accepted	Security owner

Numeric targets and fleet scale must be approved before formal acceptance. No service-level or performance commitment is inferred from these objectives.

3. Scope

3.1 Included

- Vue workshop frontend at `/workshop` and Frappe Desk for full document administration.
- Fleet master records, driver/custodian associations, meter readings and vehicle histories.
- Repair orders, operations, service templates, checklists, quality controls and linked ERPNext documents.
- Issues, faults, recalls, inspection templates, inspection records and schedules.
- Fuel quotas, refueling records, approval-state handling, expenses, parts and invoice visibility.
- Reports, saved report configurations, scheduled report execution, application settings and installation health audit.
- Hardware configuration, raw/parsed preview, active API selection, IMEI mapping, manual tests/sync, automatic polling and telemetry history.
- Docker image and service override design supplied in the repository.

3.2 Exclusions and Boundaries

- Hardware manufacture, firmware, SIM/network service, sensor calibration and vehicle ECU interpretation are provider responsibilities.
- Device-side primary/fallback POST destination selection is not controlled by this application. A single ingest API may be exposed through operator-managed redundant ingress, but that is not application-level automatic failover.
- Automatic health-ranked provider switching, offline browser operation, automated driving/remote engine control and guaranteed real-time tracking are not delivered claims.
- Customer self-service portal, bespoke government identity/payment integrations, unrelated bench apps and a complete ERP implementation are not assumed included.
- Certification decisions, legal interpretation, formal security testing and final hosting approval are outside the documentation preparation scope.

4. Stakeholders and Actors

Actor	Operational responsibility	Access expectation
Commission sponsor/product owner	Approves scope, priorities and acceptance	Governance; not necessarily a system administrator
Fleet manager	Fleet records, assignments, maintenance oversight	Approved vehicle and history access
Service advisor/workshop supervisor	Intake, planning, work progression, handover	Repair and customer/vehicle context
Technician/inspector	Assigned work, inspections, findings	Minimum records needed for assigned work
Stores officer	Parts and material movements	Inventory permissions
Fuel approvers	Department/depot approval decisions	Independently enforced approval authority required
Finance officer	Expenses, invoices and reconciliation	ERPNext finance permissions
Telemetry Integration User	Device/API ingestion service account	Telemetry create/read/list only at custom API role layer
Telemetry Integration Manager	Provider configuration and device mapping	Integration administration plus relevant Vehicle permissions
System Manager	User, schema and system administration	Privileged and audited access
Infrastructure/security operators	Hosting, backup, deployment, assurance	Controlled shell and infrastructure access

These are business personas. Not all persona labels are provisioned Frappe roles. Actual per-DocType roles are enumerated in the TDD. Legacy custom API permission gaps are recorded in the Security document; this table is not evidence of complete least-privilege enforcement.

5. Functional Requirements

5.1 Fleet and Responsibility

ID	Requirement and acceptance condition	Status / implementation
FR-FLT-01	Store vehicle identity, license plate, make/model, specifications, ownership and status; retrieve the same saved record	Implemented: Vehicle extensions and vehicle frontend
FR-FLT-02	Track driver/custodian associations and retain removed-assignment history; reject conflicting active driver assignment according to validation	Implemented: Vehicle hooks and assignment APIs
FR-FLT-03	Present service, repairs, inspections, issues, fuel, attachments and financial context on a vehicle page	Implemented: vehicle dashboard and tabs

ID	Requirement and acceptance condition	Status / implementation
FR-FLT-04	Support meter and cost histories and fleet aging/replacement analysis; distinguish advisory analytics from approved decisions	Implemented: analysis APIs and pages
FR-FLT-05	Maintain linked asset/acquisition and insurance context when required ERPNext master data is configured	Partial: setup-dependent Asset integration; validate account configuration

5.2 Workshop and Quality

ID	Requirement and acceptance condition	Status / implementation
FR-WO-01	Create a draft RO for a Company or Customer and a Vehicle, with problem, priority, operations and parts	Implemented: Repair Order controller and forms
FR-WO-02	Validate party selection, billable/FoC exclusions and SLA inputs; submit into Scheduled status	Implemented: RO validation and submit hooks
FR-WO-03	Start scheduled work, create linked Project/Tasks as needed and assign operations	Implemented: `start_work` and operation actions; submit alone does not create tasks
FR-WO-04	Track work and costs from task, timesheet, purchase/sales document events	Implemented: document hooks and cost recomputation
FR-WO-05	Generate quotation/material request and continue supported ERPNext commercial flows	Implemented: RO commercial actions, ERPNext workflow permissions still apply
FR-WO-06	Prevent Ready for Handover when a QC-linked task is incomplete; display checklist/inspection results	Implemented: submitted-order QC guard and checklist APIs
FR-WO-07	Close only according to approved financial policy	Partial: current guard accepts linked invoice status Paid or Submitted and only checks when an invoice is linked; stricter settlement policy requires review
FR-QLT-01	Record issues, faults and recalls; support approval, rejection, duplicate marking and work-order conversion	Implemented: issue APIs; authorization gaps require remediation
FR-QLT-02	Maintain inspection forms, schedules, inspections and item failures; retrieve results against the vehicle	Implemented: inspection DocTypes and pages; operational scheduling acceptance required

5.3 Fuel, Finance and Reporting

ID	Requirement and acceptance condition	Status / implementation
FR-FIN-01	Maintain monthly fuel quota and refueling records; identify over-quota requests and approval states	Implemented: fuel APIs/DocTypes
FR-FIN-02	Restrict approval decisions to authorized approvers and prevent repeated consumption posting	Required: negative authorization and repeated-approval tests; current `approve_refueling` trusts a role parameter
FR-FIN-03	Record vehicle expenses and view parts/stock and sales/purchase invoices	Implemented: expense, parts and invoice modules
FR-RPT-01	Run report library filters and view/save report configurations	Implemented: report APIs and pages; review record visibility
FR-RPT-02	Execute due report schedules and maintain job-costing snapshots	Implemented: hourly/daily scheduler hooks; delivery requires working mail/ service configuration
FR-RPT-03	Restrict sensitive exports and scheduled report recipients	Required: approved information-sharing policy and recipient/access tests

5.4 Hardware and Telemetry

ID	Requirement and acceptance condition	Status / implementation
FR-HW-01	Enter provider URL, GET/POST method, key header/key, optional JSON body and response root; test without saving or ingesting	Implemented: configuration `test` action
FR-HW-02	Display raw and parsed responses and errors, excluding the submitted provider key from preview output	Implemented: fetch/parser and frontend preview; redaction has documented limits
FR-HW-03	List saved configurations, update them and select one enabled active default per site	Implemented: configuration API and controller serialization
FR-HW-04	Associate one IMEI with one Vehicle; prevent duplicate mapping; allow explicit API or follow Active API	Implemented: unique field, Vehicle validation and connection UI
FR-HW-05	Allow authorized read-only GPS/fuel tests and manual sync for the saved enabled mapping	Implemented: `vehicle_hardware` actions
FR-HW-06	Receive provider-shaped authenticated telemetry; reject unmapped or disabled vehicles and invalid payloads	Implemented: central telemetry API; record/device scope concerns remain
FR-HW-07	Interpret GPS, speed, odometer, engine and fuel with explicit units; do not invent percentages without capacity	Implemented: shared parser and ingestion

ID	Requirement and acceptance condition	Status / implementation
FR-HW-08	Configure Running/Idle/Off intervals in whole seconds, 1-86400; defaults 10/60/120	Implemented: validation and configuration UI
FR-HW-09	Poll eligible mapped vehicles, share due-provider requests, avoid unchanged poll snapshots, record errors and retry	Implemented: long-queue polling module; timing is best effort
FR-HW-10	Show last/next request, engine condition, provider error and stored measurement timestamp; refresh visible dashboard data	Implemented: Vehicle hardware panel and five-second stored-data refresh
FR-HW-11	Preserve newer live vehicle location/ odometer against older telemetry; separate transport success from sensor freshness	Implemented guard; future timestamps and clock skew need policy/testing
FR-HW-12	Recover ingress/provider outages using the approved redundancy design	Partial: device/operator retry and configurable provider selection; automatic multi-provider failover not implemented

6. Data and Interface Requirements

ID	Requirement	Acceptance evidence
DR-01	Use IMEI text for association, not the provider's display name or registration-like label	Duplicate/unmapped-device tests and saved mapping
DR-02	Keep measurement timestamps and batch/source traceability with telemetry records	Read back a synthetic create batch and its child records
DR-03	Preserve consistent units: km, km/h, meters, degrees, liters/milliliters and percentage	Parser tests and independently calibrated provider comparison
DR-04	Define retention for location history, employee associations, financial documents, files, logs and backups	Commission-approved retention matrix; no universal automatic purge is asserted
IR-01	Serve workshop and Desk through controlled HTTPS; use authenticated Frappe APIs	Final ingress configuration and login/ API negative tests
IR-02	Integrate ERPNext vehicle, employee, customer/company, project/task, inventory and finance records	Cross-document workflow tests on the clean deployment
IR-03	Permit only approved external provider/map/mail destinations	Egress policy, provider agreement and map-data disclosure review

The provider's `dt_tracker` is preferred over `dt_server`. Incoming provider timestamps do not establish a timezone contract themselves; the integration owner must agree the timezone and stale/future tolerance. TDD section 4 defines the sample contract.

7. Nonfunctional Requirements

These are production acceptance requirements, not measured service claims.

ID	Requirement	Target / validation
NFR-SEC-01	Enforce server-side authentication, role and record-scope checks for every protected API	No unauthorized read/write in signed access matrix; Security register closure
NFR-SEC-02	Protect secrets and data in transit/at rest and prevent internal-network provider requests	HTTPS and egress controls, key management, scans and penetration tests
NFR-REL-01	Continue controlled operation after provider outage; record failure without losing other vehicles' data	Timeout/error/recovery and multi-vehicle tests
NFR-REL-02	Meet approved availability, RPO and RTO	TO BE CONFIRMED; restore rehearsal and outage exercise
NFR-PERF-01	Meet UI/API response and telemetry lag targets at agreed fleet/concurrency scale	TO BE CONFIRMED; representative load test, p95 latency, queue lag and database growth
NFR-OPS-01	Reproduce install/upgrade with one approved Docker release and migration per site	Image digest, final Compose, clean install/upgrade and audit reports
NFR-OBS-01	Alert on failed requests, stale sensors, queue backlog, storage and authentication anomalies	Monitoring demonstration; current app status fields alone are not alerting
NFR-USR-01	Support approved browsers, mobile layouts, language and keyboard workflows	Commission UAT; locale files do not establish translation completeness/ accessibility compliance
NFR-MNT-01	Version source, dependencies, documents and changes with rollback procedures	Release manifest, SBOM, review and restore evidence

Polling interval is not end-to-end freshness. Overall age includes sensor sampling, device upload, provider processing, poll interval, network/worker latency and browser refresh. An engine transition is discovered only on the next contact when no push is received.

8. Acceptance and Traceability

Acceptance ID	Requirements	Procedure and expected result
AT-01	FR-FLT-01..05	Create fictional vehicle, assign responsibility, update meter, inspect all tabs and verify linked asset conditions
AT-02	FR-WO-01..07	Draft, submit, start, complete operations, exercise QC failure/pass, create documents and test financial closure policy
AT-03	FR-QLT-01..02	Driver issue, custodian review, work-order conversion, inspection failure and follow-up; test unauthorized actor

Acceptance ID	Requirements	Procedure and expected result
AT-04	FR-FIN-01..03	Within/over-quota refueling, role denial, repeated approval and expense/invoice reconciliation
AT-05	FR-RPT-01..03	Run filtered report, save/schedule and verify delivery plus forbidden-recipient behavior
AT-06	FR-HW-01..05	Unsaved preview creates no configuration/telemetry; save lists row; switch active; map IMEI; test then sync
AT-07	FR-HW-06..11, DR-01..03	Synthetic push; wrong IMEI, disabled vehicle, invalid GPS, missing time, optional fuel, duplicate and out-of-order cases
AT-08	FR-HW-08..12, NFR-REL-01	Running/idle/off transitions, outage/recovery, worker restart, shared request, queue contention and selected-provider change
AT-09	NFR-OPS-01, NFR-MNT-01	Fresh Docker install and upgraded second site, asset HTTP checks, schema audit, database/files/key restore
AT-10	NFR-SEC-01..02, IR-01..03	Role/record negative tests, private-host request checks, TLS/secrets/image review, independent security assessment
AT-11	NFR-PERF-01, NFR-OBS-01, NFR-USR-01	Approved load/browser/language/accessibility checks and alert demonstrations

Focused parser/polling tests do not replace these end-to-end acceptance exercises. Execution evidence and current test limitations are in `verification.md`; technical source traceability is in the TDD inventory.

9. Dependencies, Risks and Sign-Off

Dependencies include ERPNext v15, compatible Frappe/runtime services, correctly configured Company/accounts/warehouses, provider availability and calibration, DNS/time synchronization, managed workers/scheduler, and an approved operating network. Demo data seeded by a new install must not become Commission operational data.

Priority risks are legacy API authorization, financial approval integrity, unverified Docker build/restore, secret/network exposure, unknown operating scale, and clock/retention policy. The Security document owns the detailed register; the SDD owns infrastructure assumptions. A feature is not accepted merely because a UI control is present.

Product reviewer: TO BE CONFIRMED. QA acceptance record: TO BE CONFIRMED. Commission approval/signature/date: TO BE CONFIRMED. Any exception must identify its owner, compensating controls and expiry; it must not be silently removed from the signed dossier.